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INTRODUCTION AND FOSS CORE CONCEPTS

Understanding Software Freedom and Open Source Fundamentals

Presented by **Nireeksha A P** · Sapthagiri NPS University.
Hosted by **SLUG** (Sapthagiri Linux Users Group)

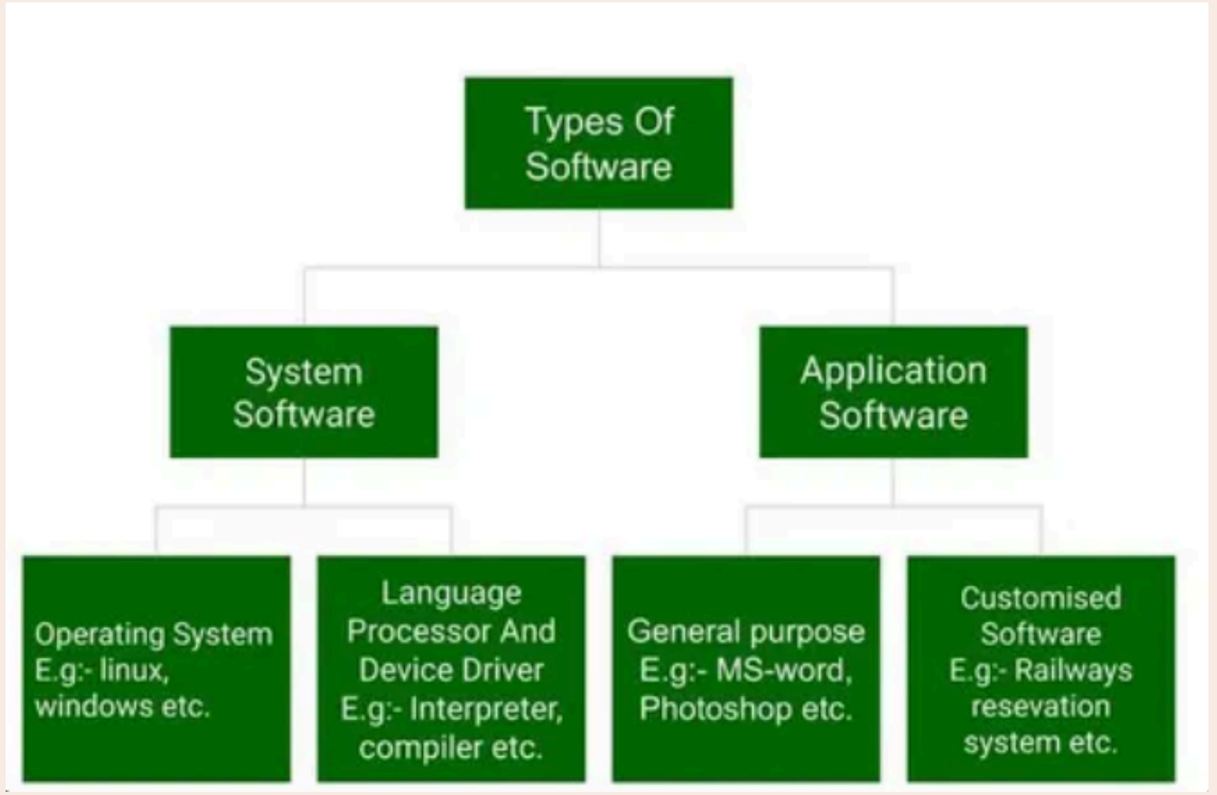
Topics TO BE COVERED

- What is Software?
- Proprietary vs FOSS
- Free Software
- Open Source Software
- Key Differences

What is Software?

Software is a collection of instructions, data, or programs that tell a computer how to work.

- **Two fundamental categories:** System Software (operating systems) and Application Software (browsers, word processors)
- **Ubiquitous presence:** Software powers smartphones, cars, medical devices, and smart homes
- **Massive industry:** The global software market is valued at over $650 billion



Proprietary vs FOSS

PROPRIETARY

- Closed-source — source code is hidden from users
- Vendor-controlled with restrictive licenses (EULAs)
- Users cannot modify, inspect, or redistribute the code

Examples: Microsoft Windows, Adobe Photoshop, macOS

FOSS

- Open Source — source code is publicly available
- Community-driven development with open licenses
- Freedom to run, study, modify, and distribute

Examples: Linux, Firefox, LibreOffice, Python

Key Distinction: Proprietary prioritizes vendor control · FOSS prioritizes user freedom

What is Free Software?

Defined by **Richard Stallman** and the Free Software Foundation (FSF) in 1985.
"Free" refers to **freedom**, not price.

The Four Essential Freedoms

0. **Run** the program for any purpose
1. **Study** and modify the source code
2. **Redistribute** copies to help others
3. **Distribute** modified versions to the community



Examples: GNU/Linux, Emacs, GCC (GNU Compiler Collection)

Focus: User rights and ethical principles of software freedom

What is Open Source?

Open Source emphasizes the **practical benefits** of publicly available source code.

- Coined in **1998** by the **Open Source Initiative (OSI)** as a more business-friendly term
- The **Open Source Definition (OSD)** specifies 10 criteria for certification
- Focuses on **collaboration**, **transparency**, and community-driven development
- Key criteria include free redistribution, source code availability, and license integrity

Examples: Apache HTTP Server, Python, Android, VS Code



Focus: Practical development methodology and collaboration

Free Software vs. Open Source

Dimension	Free Software (FSF)	Open Source (OSI)
Philosophy	User freedom and ethics	Practical benefits and methodology
Terminology	"Free Software" — freedom-focused	"Open Source" — business-friendly
Legal Base	GPL, Copyleft licenses	MIT, Apache 2.0, BSD
Priority	User rights and control	Collaboration and innovation



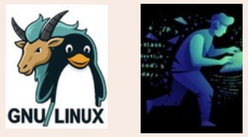
Key Insight: All Free Software is Open Source, but not all Open Source is Free Software (philosophically). The combined term **"FOSS"** bridges both movements.



Key Takeaways

- **"Free"** in FOSS means **user freedom** and data control, not zero cost
- **"Open Source"** means the code blueprint is publicly available for anyone to inspect
- FOSS empowers users with the **four essential freedoms**
- Understanding the difference helps you choose the right software and licensing

Thank you for your attention

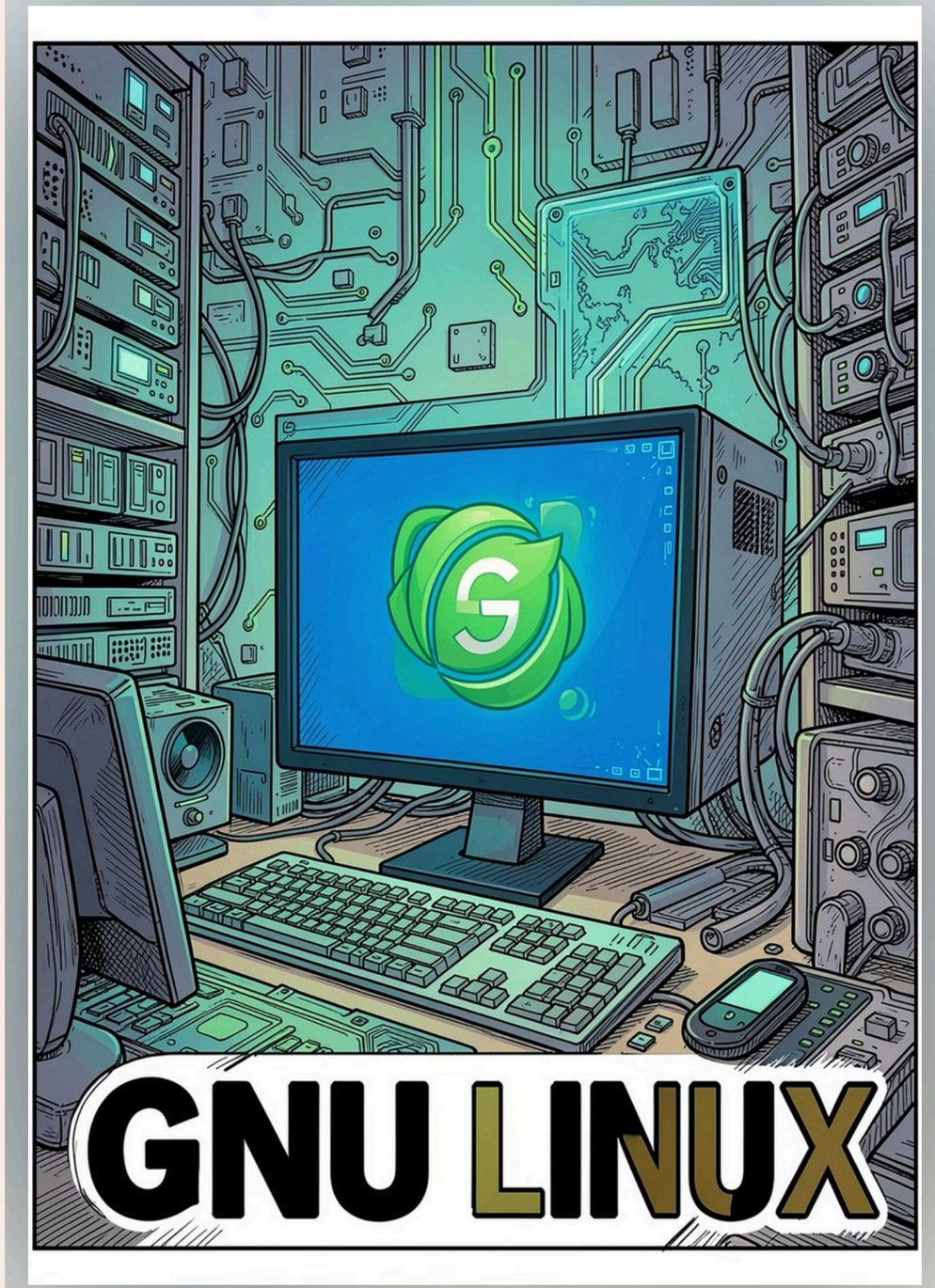


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INTRODUCTION TO GNU LINUX

Exploring the OS Behind the Modern Internet & Career Growth

Presented by **Omkar Udas Thakarkar -** Sapthagiri NPS University
Hosted by **SLUG** (Sapthagiri Linux Users Group)



The Digital Bridge

An operating system is the critical mediator between human intent and machine hardware — the invisible layer that makes every digital action possible.

Resource Control

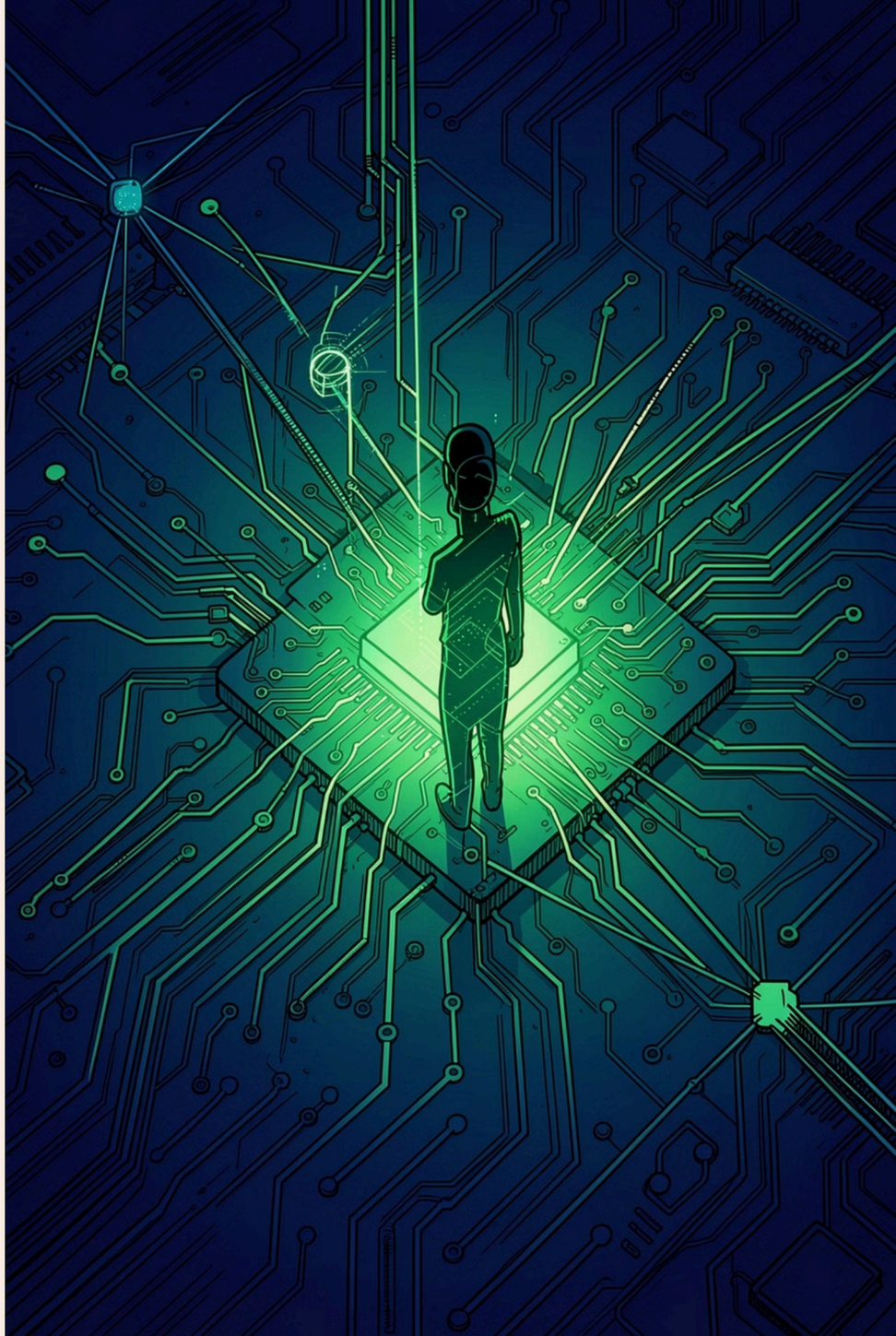
Dynamically manages CPU allocation and memory distribution across all running processes.

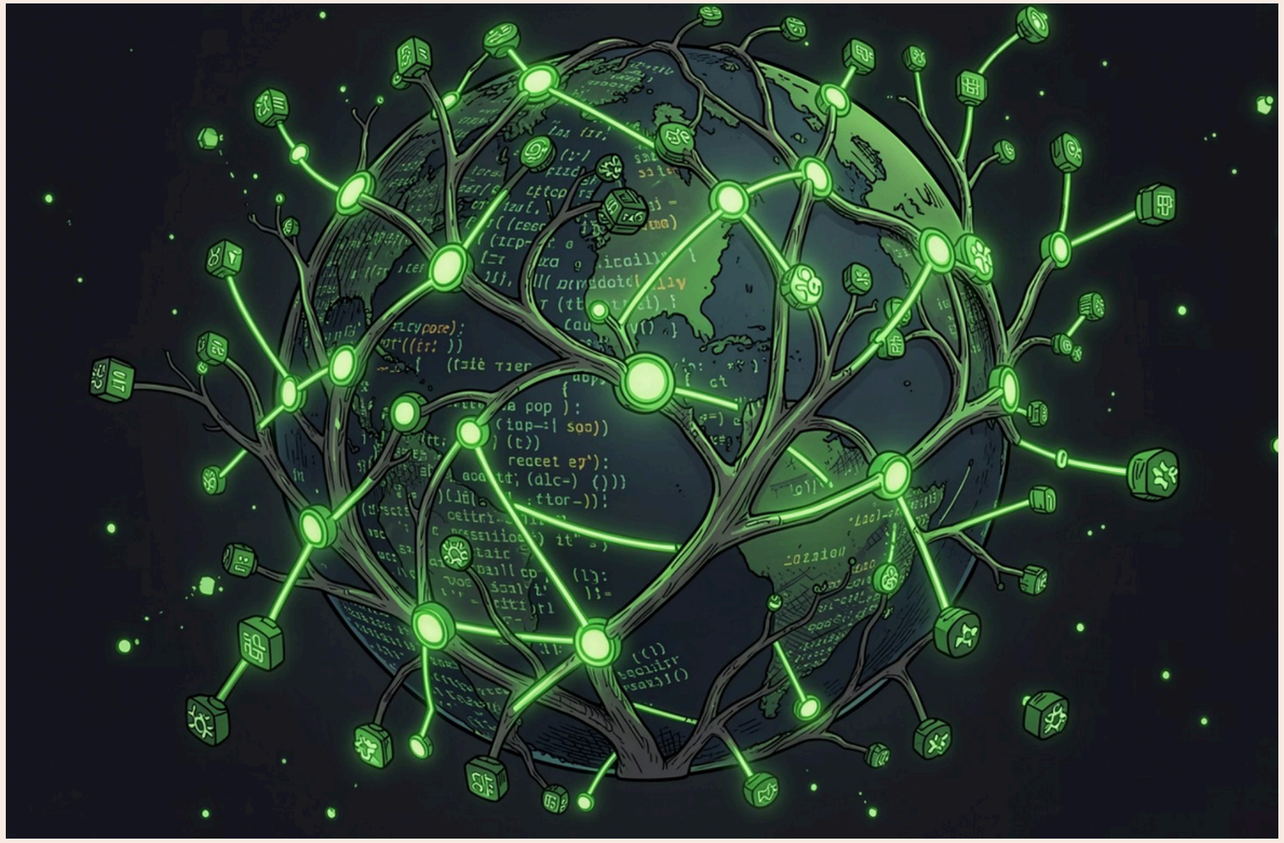
Data Organization

Structures the directory tree, storage volumes, and file networks for seamless access.

Device Integration

Standardizes keyboard, mouse, and hardware inputs into a unified interface.





FOUNDED 1991

The Open-Source Revolution

Created by **Linus Torvalds**, Linux is a globally developed kernel — built collaboratively, freely shared, and continuously improved by thousands of contributors worldwide.

→ **Pure Open Source**

Anyone can inspect, modify, secure, and contribute to the source code.

→ **The Distribution Ecosystem**

Packaged into diverse "Distros" tailored for every use case.

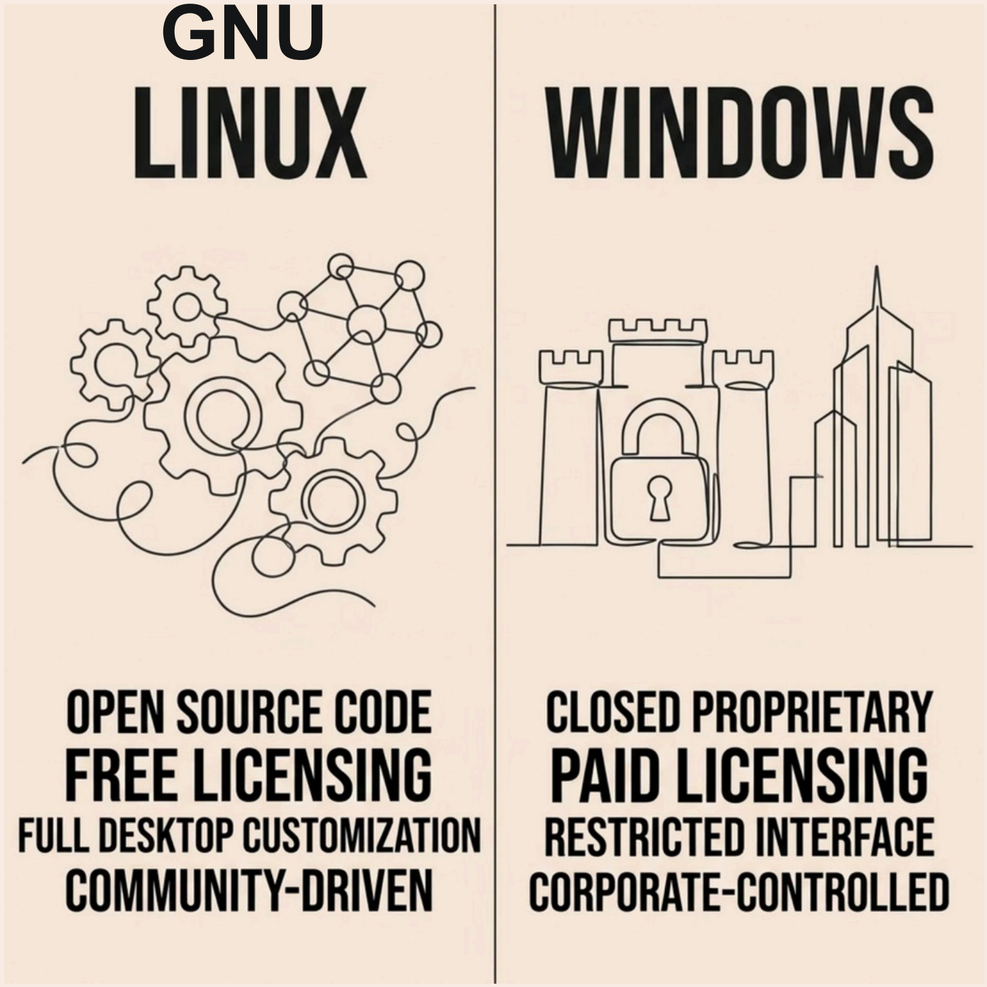
→ **Famous Flavors**

Ubuntu(beginner-friendly), Fedora(developer-centric), Arch Linux (power users).

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Open vs. Closed Ecosystems

GNU Linux and Windows represent fundamentally different philosophies — one built on transparency and community, the other on control and commerce.



Three Core Distinctions

Open vs. Proprietary

GNU Linux code is transparent and auditable; Windows is closed-source and strictly controlled by Microsoft.

Licensing Cost

GNU Linux is 100% free and open-source; Windows requires paid commercial licensing.

Desktop Customization

GNU Linux lets users alter every pixel and workflow; Windows restricts system interface customization.

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Engineered for Efficiency

Lightweight Footprint

Low resource usage revives old hardware that crawls under Windows — extending device lifespan and reducing e-waste.

Security by Design

Strict root vs. user permission walls minimize unauthorized execution and privilege escalation attacks.

Ransomware Resiliency

Far fewer security vulnerabilities and targeted malware attacks in standard Linux environments.

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COMMAND LINE INTERFACE

The Power of the CLI

The terminal is a fast, precise, and highly scriptable alternative to clicking through a GUI. One line can replace minutes of manual work.

```
mkdir project
```

Create a project directory instantly.

```
sudo apt install git
```

Install developer software in one command.

```
ls -la
```

List all hidden and system files.

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The Invisible Powerhouse

GNU Linux is the quiet engine behind modern non-technical industries — from consumer privacy to hardware sustainability.

Hardware Revitalizer

Saves 10-year-old consumer laptops from electronic landfills by breathing new life into aging hardware.

Ultimate Privacy

Zero background telemetry, ad-trackers, or forced OS updates — your machine stays yours.

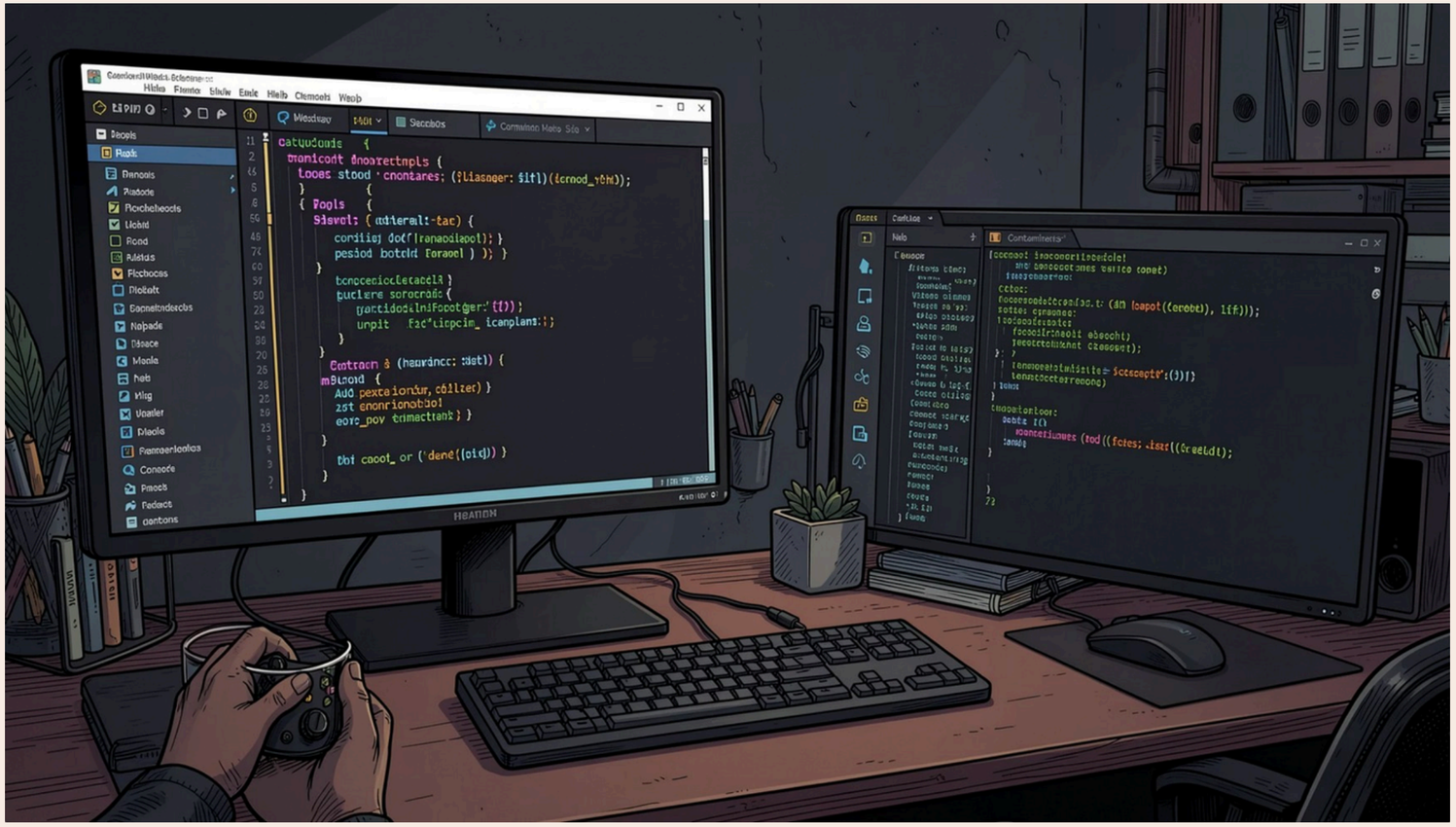
Smarter Computing

The ideal foundation for home media servers, custom network storage, and personal cloud infrastructure.

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The Developer's Native Tongue

For engineering students, Linux isn't just an OS — it's the environment where professional-grade development naturally.



1 First-Class Code Support

Native, hassle-free interpreters and compilers for Python, C/C++, Java, and Rust.

2 Toolchain Hegemony

Docker containers, Git workflows, and package managers work natively and seamlessly.

3 System Comprehension

Forces you to understand how directory trees, file configurations, and processes operate under the hood.

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The Guardian's Environment

Industry-standard cybersecurity software is natively engineered for Linux. Professionals in ethical hacking and network defense rely on it exclusively.

Kali Linux

The world-standard platform optimized for professional network penetration testing and security auditing.

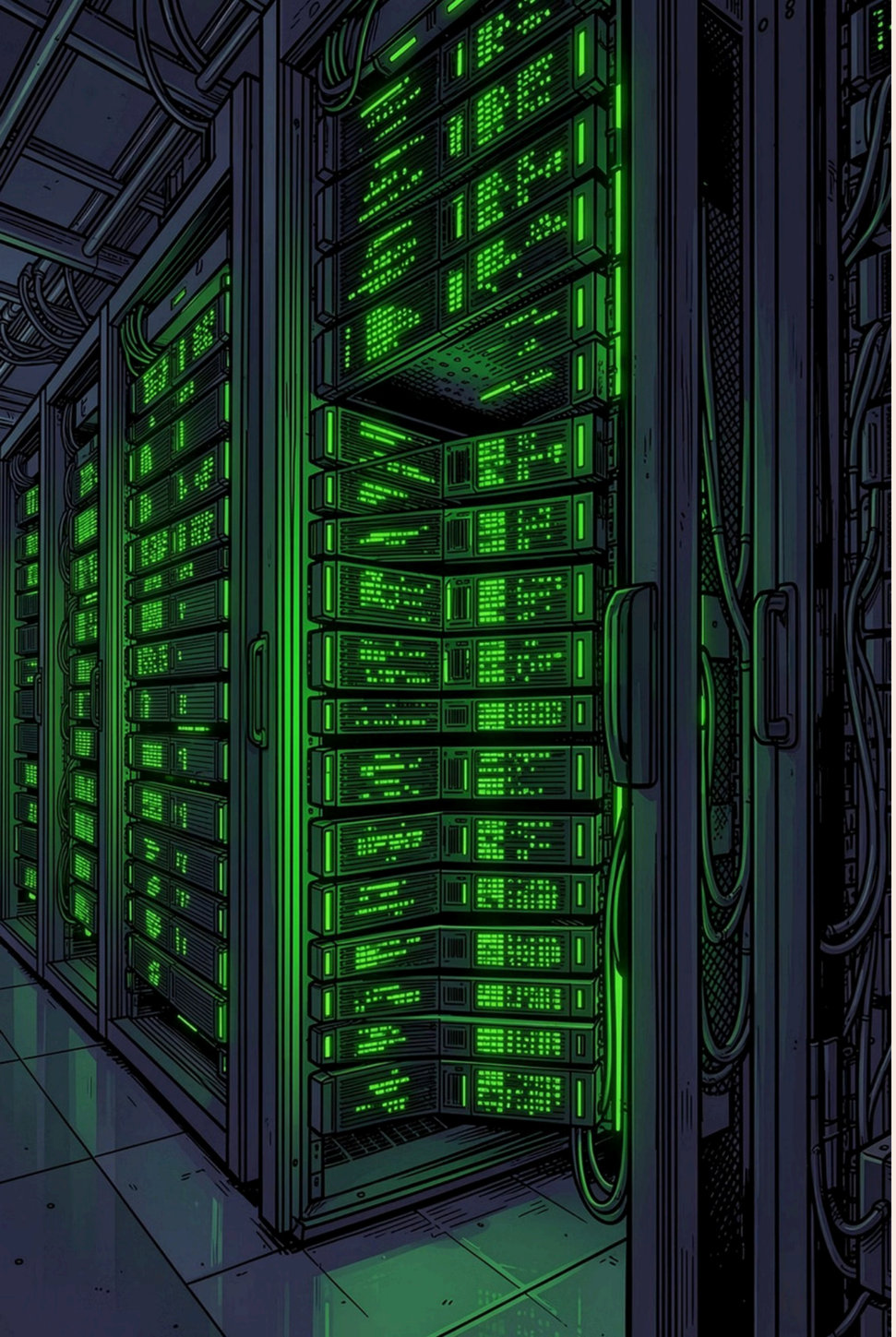
Under-the-Hood Access

Direct control over low-level networking, network interface cards, and kernel drivers for deep analysis.

Built-In Tools

Native integration of Wireshark, Nmap, and dozens of diagnostics and mapping programs out of the box.

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The Enterprise Advantage

100%

Top 500 Supercomputers

Every one of the world's most powerful supercomputers runs on Linux.

#1

Cloud & DevOps Standard

AWS, Azure, and AI roles require command line fluency as a baseline skill.

3B+

Android Devices

Android — the world's most popular mobile OS — runs on a modified Linux kernel.

SpaceX Falcon 9 rockets, Dragon capsules, and NASA hardware all run mission-critical operations on Linux. The vast majority of global public clouds and enterprise databases run exclusively on Linux.

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